NEEL BANSAL · ANDERS HAROLDSON



HACKHARVARD 2026 · SPACEXAI CHALLENGE

TEARDOWN

BRINGING ROCKET SCIENCE DOWN TO EARTH

THE PROBLEM

Millions watch Falcon 9 launch. Almost nobody can explain what they saw.

The facts are all public. They are scattered across spec sheets, webcast timestamps and launch databases.

A webcast is something you watch. You can't pull it apart, ask it a question, or try to fly it yourself.

So rocket science stays something other people understand.



NO WAY TO TOUCH IT

Three places, three formats, nothing you can hold.

OUR MOTTO

BRINGING ROCKET SCIENCE DOWN TO EARTH

Teardown puts a real Falcon 9 in one place you can touch.

Take it apart. Watch a real launch. Then slide a phone into a cardboard rocket and fly your own flight to orbit.

Grok explains every moment, in words a 10-year-old understands.

3
WAYS TO PLAY

10
PARTS TO PULL APART

3
READING LEVELS

1
PHONE FOR A ROCKET

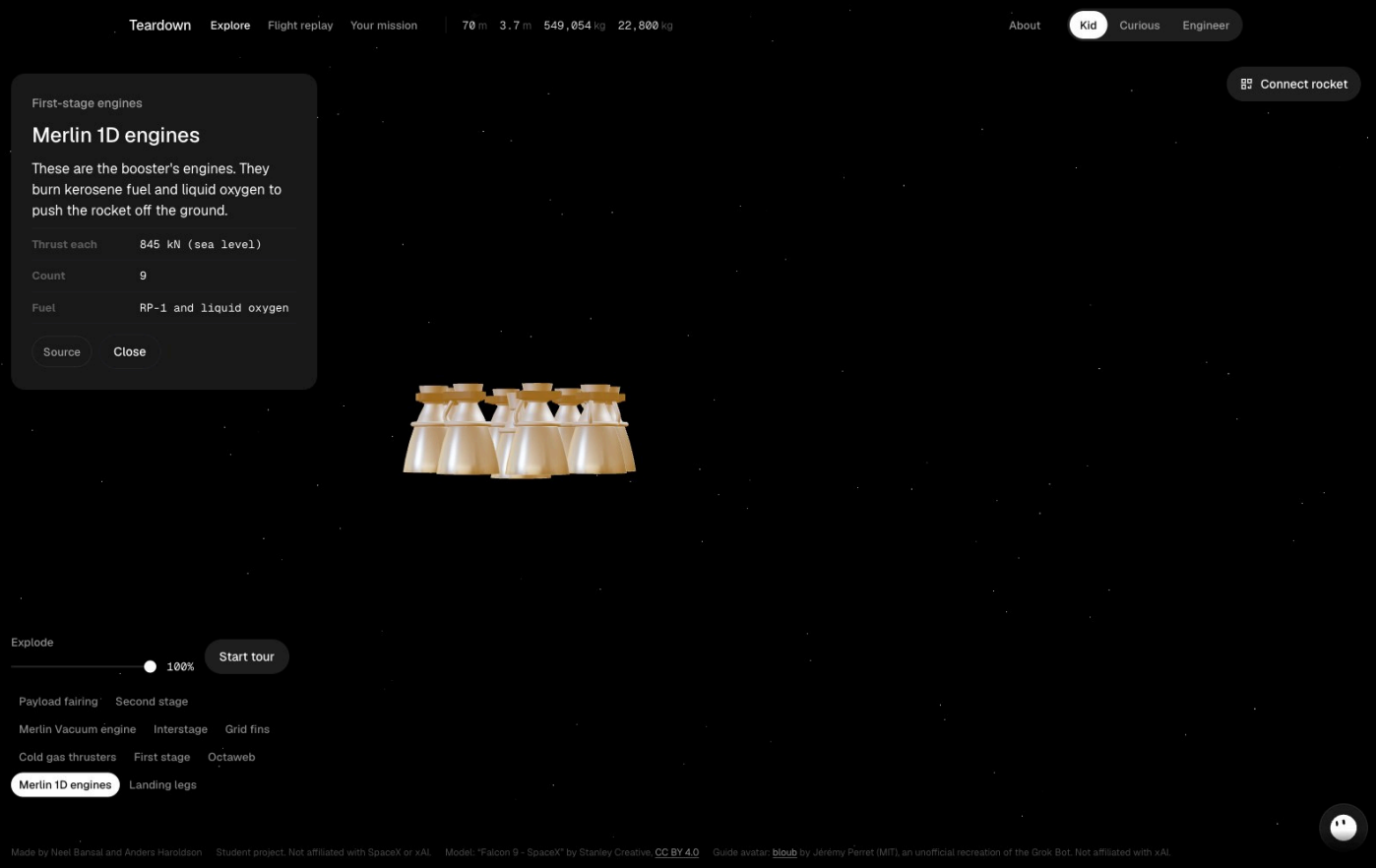
01 · EXPLORE

TAKE IT APART

Drag to turn a Falcon 9. Slide Explode and it splits into its 10 real parts, from the fairing to the landing legs.

Tap a part: the camera flies to it, and a card explains it with real specs and a source link.

Kid, Curious, Engineer. Same facts, more depth.



The running app. Exploded, Merlin 1D engines selected.

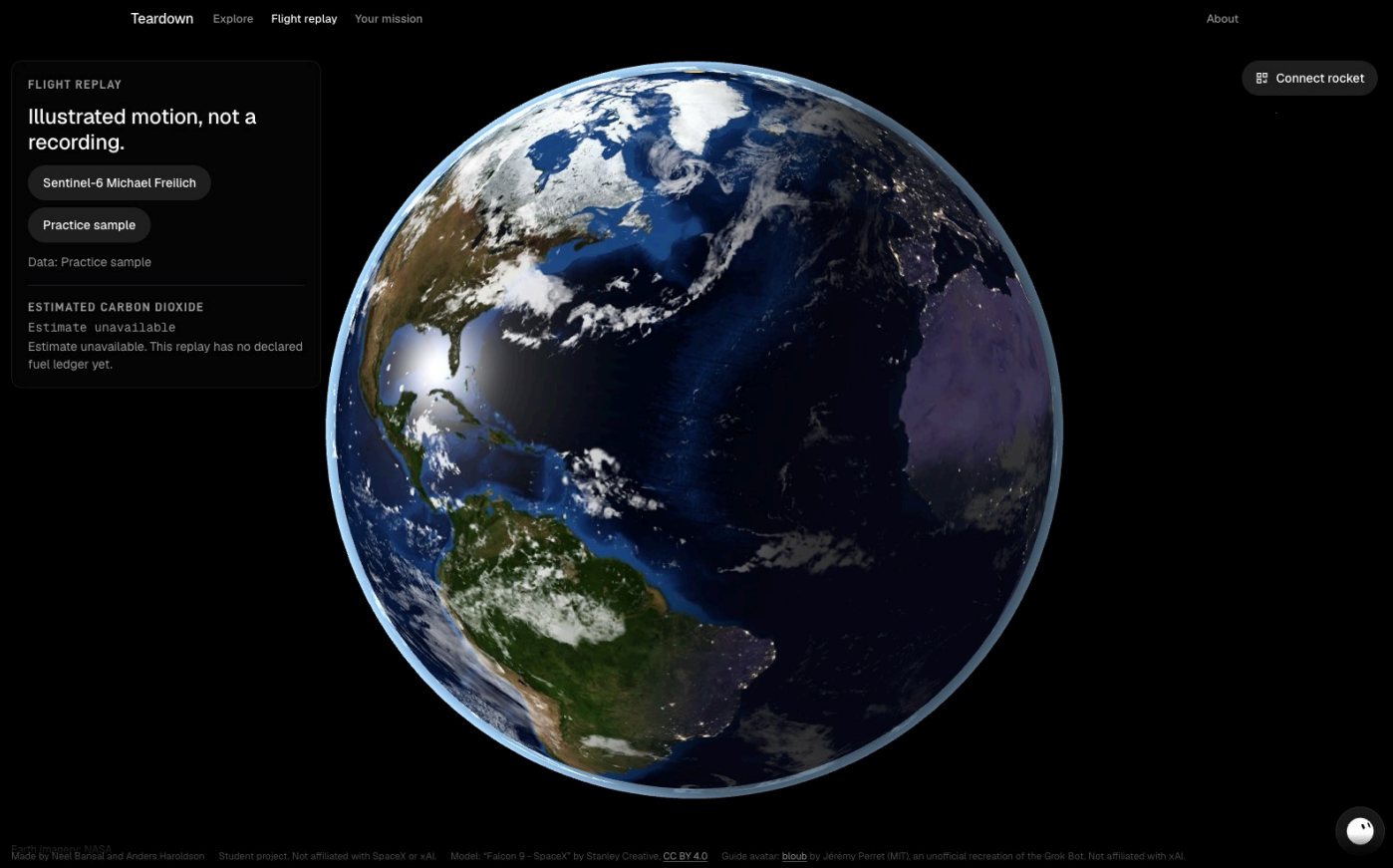
02 · FLIGHT REPLAY

WATCH A REAL LAUNCH

Pick Sentinel-6 Michael Freilich, Falcon 9, 21 November 2020.

The rocket flies around a NASA Blue Marble Earth and hits each milestone at its published time.

It says plainly what it is: illustrated motion, not a recording, and which numbers the public sources leave out.



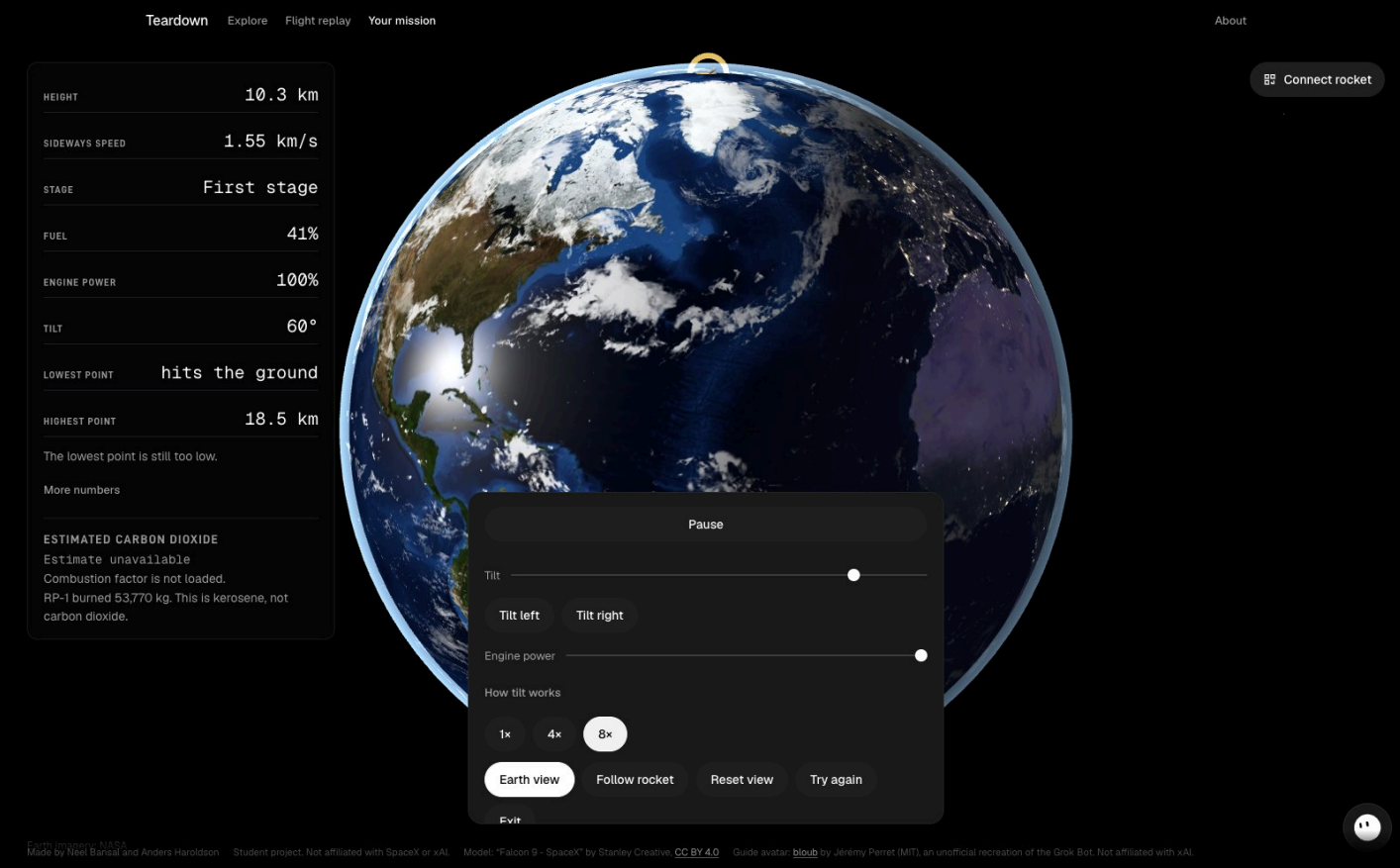
The running app. NASA Blue Marble Earth, Sentinel-6 picker.

FLY IT TO ORBIT

You control tilt and engine power. Real physics does the rest, and nothing types in a win.

> **Go straight up and you fall back. Turn too hard and you hit the ground. That is the lesson of orbit, learned by failing.**

Win: lowest point at least 200 km up, highest no more than 2,000 km, engines off for ten seconds.



The running app. Height, sideways speed, stage, fuel, lowest and highest point, live.

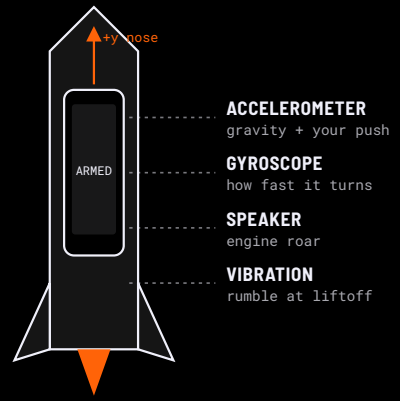
YOUR PHONE IS THE ROCKET

A phone slides into a cardboard rocket and becomes its motion sensor and its speaker. No app to install, just a web page.

Scan the QR on the laptop. Tap Start. Hold it upright to zero. Then lean the rocket to steer.

It uses the phone's accelerometer and gyroscope for motion, and its speaker and vibration motor for engine sound and rumble.

> **We checked with the HackHarvard co-director: the phone's accelerometer and speaker count as hardware for this track.**



1 SCAN · 2 ALLOW · 3 ZERO · 4 TILT

IN DEPTH · SENSORS

From raw motion to a steering angle

The phone reads devicemotion and deviceorientation, and sends a frame at most every 50 ms, about 20 Hz.

It splits acceleration into gravity and your movement. Gravity is low-passed, so a shaky hand doesn't jerk the rocket.

Zero: hold it still for 2 seconds. That averaged gravity becomes "straight up", so any phone in any mount works.

Tilt is gravity in the rocket's frame: pitch = atan2(g_z, √(g_x² + g_y²)), clamped to ±90°.

PHONE SENSORS	`devicemotion + deviceorientation · ≤ 20 Hz`
SPLIT	`gravity (low-pass 0.8) | your movement`
ZERO	`2 s still → rocket frame, +y = nose`
PITCH	`atan2(g_z, √(g_x² + g_y²)) · clamp ±90°`
RELAY → LAPTOP	`{q, lin, grav} · Supabase Realtime · 4-letter code`
STORE → PHYSICS	`one owner per control · pause if silent 2 s`

IN DEPTH · MOTION CONTROLS

The cardboard rocket is the controller

Every gesture is measured in the rocket's own frame, along the calibrated nose, not the phone's screen.

Each one needs a sustained or shaped signal, so a table bump or one noisy sample never launches.

If no frame arrives for 2 seconds, the phone gives up control and the flight pauses. Stale input is thrown out.

MOVE	WHAT THE SENSOR MUST SEE	DOES
LIFT	≥ 2.5 m/s² up the nose for 180 ms	**Launch**
LEAN	gravity angle in the rocket frame	**Steer**
PUSH	≥ 3 m/s² along the nose	**Engine power**
FLICK	≥ 12 m/s² sideways, 40–220 ms	**Drop stage 1**
NOSE DOWN	past 150° for 2 s	**Abort**
SILENCE	no frame for 2 s	**Pause**

Thresholds from lib/rocket/shell-gestures.ts.

IN DEPTH · PHYSICS

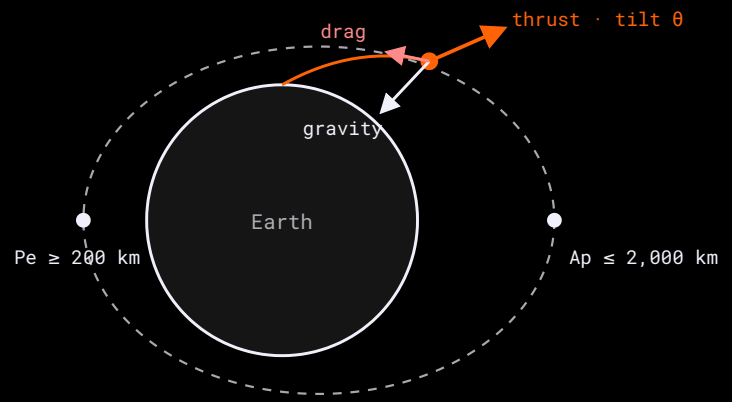
How we compute where the rocket is

Every 1/60 of a second we add up the forces: gravity toward Earth's centre, engine thrust along your tilt, and air drag.

Then update speed, then position. Mass drops as fuel burns, so the rocket speeds up as it gets lighter.

From position and speed we compute the whole orbit live: its energy, its shape, its lowest and highest points.

The win is checked on that orbit, never typed in. Time warp at 4× or 8× just runs more steps per frame.



$$a = -\mu \cdot \hat{r}/r^2 + (T/m)(\cos\theta \cdot \hat{r} + \sin\theta \cdot \hat{t}) - D/m$$

$$v \leftarrow v + a \cdot dt \qquad x \leftarrow x + v \cdot dt \qquad dt = 1/60\ \text{s}$$

$$\varepsilon = v^2/2 - \mu/r \qquad h = x \cdot v_y - y \cdot v_x$$

$$e = \sqrt{1 + 2\varepsilon h^2/\mu^2} \qquad Pe,\ Ap = a(1 \mp e) - R_\oplus$$

win: Pe ≥ 200 km · Ap ≤ 2,000 km · 10 s coast

ASK IT ANYTHING

Grok can select parts, explode the rocket and fly the camera by calling the same actions the buttons use.

Grok Voice speaks the guide and the mission callouts. Hold to talk and Grok transcribes your question.

Grok can explain a flight. It can't steer one or award a win.

AGENT

Function calling. Its tools are the store actions the buttons use.

VOICE

Speaks the guide and callouts. Hold to talk, Grok transcribes.

IMAGINE

"Show in action" draws the selected part, grounded in its data.

GROK BOT

A corner face that reacts while Grok listens and talks.

The xAI key stays on the server. If Grok is unreachable, the exhibit keeps running.

HOW IT WORKS

ONE STORE.
THREE WAYS IN.

Nothing moves the scene except store actions. The phone, the mouse and Grok all call the same ones.

Store actions are Grok's tools. A test keeps the two lists in sync.

Facts live in data files. Every number on screen or spoken aloud is read from data/.

About 700 unit tests, plus a smoke run that pairs a real laptop page with a phone page and flies.



SPACEXAI CHALLENGE

BUILT FOR THIS TRACK

Every requirement is met in the running app, not on a roadmap.

✓ **REAL SPACE DATA**	Falcon 9 specs from spacex.com, the Sentinel-6 launch timeline, NASA Earth imagery, a sourced emissions file
✓ **GROK VOICE OR IMAGINE**	Both, plus Grok function calling that drives the scene
✓ **BUILT WITH CURSOR**	Built in Cursor, plans in docs/plans/
✓ **HARDWARE FOR GOOD**	A phone in a cardboard rocket: accelerometer, gyro, speaker. Confirmed with the co-director

TRY IT

Take one apart. Watch one. Fly one.

BRINGING ROCKET SCIENCE DOWN TO EARTH.

Come fly the cardboard rocket. Thank you.

Made by Neel Bansal and Anders Haroldson

Student project. Not affiliated with SpaceX or xAI.

Model: “Falcon 9 - SpaceX” by Stanley Creative, CC BY 4.0. Model page

EXPLORE REPLAY FLY

APPENDIX · NUMBERS

Where every number comes from

A published fact wears its source: spacex.com, ESA, NASA, Launch Library 2.

A simulated readout comes from the physics and your inputs.

An estimate wears that word and its assumptions, like the CO_2 factor.

Missing data stays missing. We never fill a gap with a made-up flight.

SOURCE
SIMULATION
ESTIMATE
MISSING STAYS MISSING

APPENDIX · STATUS

What is still open

As of 17 October 2026, Explore, Flight replay, Your mission and the phone link run on main.

- [ ] The phone rocket has run against the phone simulator and the paired smoke test, not yet on a real handset over HTTPS.
- [ ] Sentinel-6 has published milestone times but no public altitude or speed series, so the replay draws no corridor.